

Bachelor in Artificial Intelligence
Web and Social Media Search and Analysis



Framing AI in the Hollywood Strike: A Social Media Analysis of Online Communities

Final Project Report

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Abstract

The 2023 Hollywood Strike generated a highly visible public debate about the role of artificial intelligence in creative industries. This study investigates how Reddit communities framed AI during the dispute and whether these interpretations varied across communities and over time. Using a dataset of 399 submissions and 11,048 comments collected from five subreddits between March and December 2023, the analysis combines frame detection, sentiment analysis, named entity recognition, and network analysis. Five dominant frames were identified: Replacement Anxiety, Workers' Rights, Authenticity and Creativity, Corporate Power, and Innovation Opportunity. The results show substantial differences across communities. Technology-oriented users emphasized innovation and technological progress, while professional screenwriting communities focused on authorship, creative identity, and labor concerns. Network analysis revealed a highly segmented discussion structure, with interactions occurring mainly within communities rather than across them.

1. Introduction

The 2023 Hollywood Strike marked the first simultaneous work stoppage by the Writers Guild of America (WGA) and The Screen Actors Guild-American Federation of Television and Radio Artists (SAG-AFTRA) since 1960. At the center of both disputes was **the rapid emergence of generative AI**: studios sought to use AI systems for script generation, story development, and digital actor replication, while unions demanded contractual protections against displacement and unauthorized use of members' likenesses and creative work. The strike lasted **from May to September 2023**, producing one of the most visible public debates about AI and labor in recent memory. **Reddit** became a primary venue for this debate. Its community structure, organized around topically distinct subreddits, makes it particularly suited for studying how different audiences interpret the same phenomenon. Communities centered on screenwriting in r/Screenwriting, technology-oriented discussions in r/artificial, and broader entertainment audiences in r/movies and r/television engaged with the same events from different vantage points, potentially constructing distinct interpretive frames around AI.

2. Objective and Research Question

This project investigates how online communities framed the role of artificial intelligence during the Hollywood Strike. Framing is understood here as the selective emphasis on particular aspects of an issue that shapes how audiences interpret and respond to it. Five analytical frames are identified: replacement anxiety, workers' rights, creative authenticity, corporate power, and technological opportunity.

The central research question is: **How did online communities frame the role of AI during the Hollywood Strike, and did framing vary systematically across communities and across the temporal phases of the dispute?**

This question requires integrating structural and content dimensions of analysis: identifying not only what was said about AI, but who said it, in which community, at what moment of the dispute, and whether interaction patterns reinforced or cut across the interpretive boundaries between communities.

3. Data

3.1 Collection Strategy

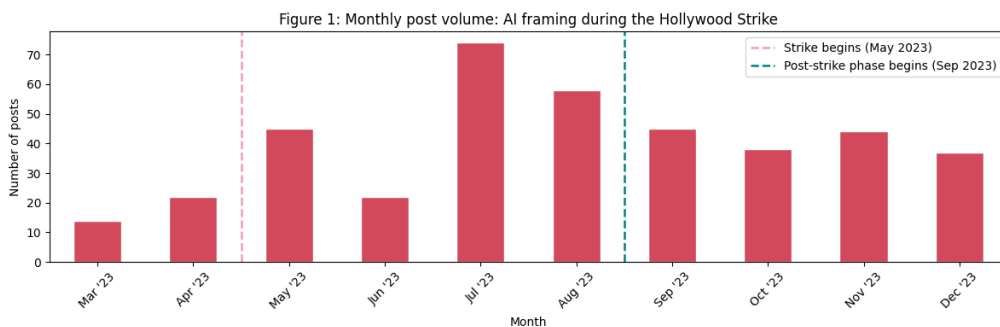
Data was collected from Reddit using the **PullPush** historical archive, which provides access to Reddit submissions and comments. Reddit was selected because it hosts large-scale public discussions organized into topic-specific communities, making it suitable for comparing how different audiences interpret AI during the 2023 Hollywood Strike.

The analysis covers the period **from March to December 2023** and distinguishes three temporal phases: **pre-strike** (March–April), **strike** (May–August), and **post-strike** (September–December). Five subreddits were selected to capture different perspectives on the debate: **r/Screenwriting**, **r/movies**, **r/artificial**, **r/television** and **r/labor**.

Data collection was guided by nineteen search queries combining broad references to the Hollywood Strike and AI with more specific themes such as job replacement, workers’ rights, creative authenticity, and contractual negotiations. For each submission, metadata including author, timestamp, engagement metrics, subreddit, title, and text were collected. Associated comments were subsequently retrieved to support network and content analyses.

3.2 Corpus Description

The final corpus consists of **399 Reddit submissions** and **11,048** associated **comments** collected between March and December 2023.



As shown in Figure 1, discussion volume increased substantially during the strike and peaked in **July 2023**, when both the WGA and SAG-AFTRA were actively involved. Although activity declined after September, discussions remained relatively intense throughout the post-strike period, suggesting that AI-related concerns continued beyond the resolution of the dispute.

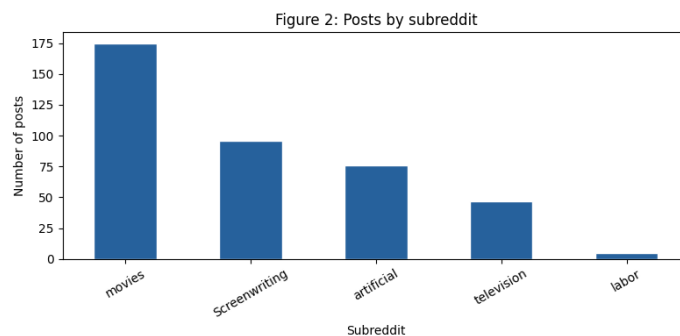


Figure 2 shows the **distribution of submissions** across subreddits. Most posts originate from **r/movies** (175), followed by **r/Screenwriting** (96), **r/artificial** (76), and **r/television** (47). Only five submissions were collected from **r/labor**, which was therefore excluded from most community-level comparisons. The **comments** dataset follows a similar pattern. Most comments were produced during the post-strike phase (6,799), followed by the strike period (3,501) and the pre-strike phase (748). The large number of comments provides a rich interaction structure for network analysis and supports the investigation of user engagement and community dynamics.

4. Methodology

4.1 Data Preprocessing

The collected Reddit corpus was cleaned and standardized before conducting content and network analyses. Social media data often contain noise, including irrelevant posts, deleted content, automated activity, non-English texts, and very short messages with limited analytical value. A preprocessing pipeline was therefore applied to both submissions and comments.

First, a **relevance filter** was used to remove posts only marginally related to AI and the Hollywood Strike. **Deleted** or **removed content** was cleaned, while **automated accounts** and obvious **bots** were excluded to ensure that the dataset reflected genuine user participation. Only English-language content was retained, as the sentiment and named entity recognition models employed later in the analysis were optimized for English text. The corpus was then normalized by removing URLs, punctuation, special characters, and Reddit-specific references, while converting all text to lowercase. Text processing was performed using **spaCy**. Documents were **tokenized** and **lemmatized**, and standard English **stopwords** were extended with a domain-specific stopword list to remove highly frequent terms that provided little analytical value. Finally, documents containing fewer than five tokens were excluded, as they lacked sufficient contextual information for reliable interpretation. The same preprocessing procedure was applied to both submissions and comments. The resulting cleaned corpora served as the input for later stages of the study.

4.2 Frame Detection

Frame detection was used to identify the main interpretive perspectives through which Reddit users discussed artificial intelligence during the Hollywood Strike. Building on the research question and the themes emerging from the exploratory lexical analysis, a dictionary-based approach was adopted to classify posts according to a set of theoretically relevant frames.

The objective was to identify the dominant narrative through which AI was interpreted, such as a threat to employment, a labor-rights issue, a challenge to creative authenticity, an expression of corporate influence, or an opportunity for innovation.

4.2.1 Exploratory Lexical Analysis

Before applying frame detection, an exploratory lexical analysis was conducted on the cleaned corpus to verify whether the expected themes emerged naturally from the data.

Word-frequency and **bigram analyses** were used to identify the most common concepts and recurring expressions. This preliminary step served as a validation procedure, confirming that discussions were centered on topics directly related to creative work, entertainment production, labor

relations, and generative AI technologies. The lexical patterns identified during this stage provided empirical support for the subsequent construction of the frame dictionaries.

4.2.2 Frame Classification Method

Five theory-driven frame dictionaries were manually constructed to capture the main dimensions of the debate. The frames represent concerns about job displacement (**Replacement Anxiety**), labor protections and collective bargaining (**Workers' Rights**), human authorship and originality (**Authenticity and Creativity**), institutional influence and profit-driven decision making (**Corporate Power**), and the potential benefits of AI adoption (**Innovation Opportunity**).

For each post, keyword occurrences associated with each frame were counted and the frame with the highest score was assigned as the **dominant frame**. Posts containing evidence for multiple frames with comparable scores were labeled as *mixed*, while posts without sufficient frame-related indicators were classified as *unclassified*. The same procedure was subsequently applied to comments, enabling later integration with network-level analyses.

4.2.3 Comparative Frame Analysis

Once dominant frames had been assigned, frame distributions were examined across multiple analytical dimensions. First, overall frame frequencies were calculated to identify the most common interpretations of AI within the corpus. Second, frame distributions were compared across subreddits to assess whether different online communities emphasized different aspects of the debate. Finally, frame distributions were analyzed across the pre-strike, strike, and post-strike phases to investigate whether framing patterns changed over time.

These comparisons provide the basis for evaluating how different audiences interpreted AI and how those interpretations evolved throughout the Hollywood Strike.

4.3 Network Analysis

Network analysis is used to examine the structure of interactions underlying discussions about artificial intelligence during the Hollywood Strike. The principal goal is to identify influential participants, interaction patterns, and conversational communities that emerged during the debate. The analysis is based on reply relationships extracted from the cleaned Reddit comments dataset. By reconstructing these interactions as a graph, it becomes possible to investigate both the overall structure of the discussion and the role played by individual users and communities within the broader conversation.

4.3.1 Network Construction

The interaction network is constructed from the cleaned comments corpus. In this network, nodes represent Reddit users and edges represent reply interactions between users. A directed edge is created whenever a user replies either to another comment or directly to the author of a submission. The network is modeled as a **directed weighted graph**. Edge direction preserves the flow of communication, while edge weights capture repeated interactions between the same pair of users. Self-replies, deleted accounts, and interactions involving missing authors are excluded from the graph construction process.

4.3.2 Centrality and Brokerage Analysis

To identify structurally important participants, several **network centrality measures** are computed. In-degree centrality captures the extent to which a user receives replies from others, while out-degree centrality measures how actively a user responds within discussions.

Particular attention is given to betweenness centrality, which measures how frequently a user lies on the shortest paths connecting different parts of the network. Users with high betweenness centrality can act as brokers between otherwise disconnected groups and may play an important role in the diffusion of information, interpretations, and narratives across the discussion space. The resulting centrality measures provide a quantitative basis for identifying influential participants and examining whether key users occupy bridging positions between communities.

4.3.3 Community Detection

To identify groups of users who interact more frequently with one another than with the rest of the network, community detection is performed using the **Louvain algorithm**. Since community detection focuses on patterns of interaction rather than communication direction, the analysis is conducted on an undirected version of the reply network.

The Louvain algorithm partitions the network into communities by maximizing **modularity**, a measure that evaluates the extent to which interactions occur within communities rather than between them. Higher modularity values indicate a more clearly segmented community structure. Community detection makes it possible to investigate whether different groups of users developed partially independent conversational spaces and whether these communities were associated with different subreddits, audiences or framing patterns.

4.3.4 Temporal Evolution of the Network

To examine how interaction patterns evolved throughout the Hollywood Strike, network analysis is repeated separately for the pre-strike, strike, and post-strike phases.

For each temporal phase, network size, community structure, and modularity are computed independently. Comparing these measures across phases makes it possible to assess whether the strike period altered the organization of online discussions by increasing fragmentation, strengthening community boundaries, or changing the overall structure of user interactions.

This longitudinal perspective allows network dynamics to be linked directly to the major stages of the labor dispute and provides insight into how public engagement evolved over time.

4.4 Content analysis

While network analysis focuses on the structure of interactions, content analysis examines the meaning conveyed through those interactions. It provides insight into the narratives, sentiments, and actors that shape online discussions. In the context of this project, content analysis is used to understand how AI was discussed during the Hollywood Strike and whether different communities adopted different interpretive perspectives toward the technology.

4.4.1 Sentiment Analysis

Sentiment analysis is performed using “*cardiffnlp/twitter-roberta-base-sentiment*”, a **RoBERTa-based transformer model** fine-tuned on approximately 58 million tweets. The choice of a

transformer model over other alternatives was motivated by the consideration that lexicon-based approaches such as VADER or AFINN assign fixed polarity scores to individual tokens without accounting for context, negation, or informal language patterns. In contrast, a transformer-based model evaluates the full contextual syntax of each input, making it substantially more appropriate for Reddit text, which shares structural properties with microblogging platforms in its use of informal register, irony, and opinion-heavy phrasing.

The model classifies each post into one of three labels: **positive**, **neutral**, or **negative**, and returns a confidence score for the assigned label. Posts exceeding the model's 512-token context window are automatically truncated, while posts with missing or empty text are assigned a neutral label with zero confidence. The confidence score is retained alongside each prediction and used as an additional quality-control indicator.

Sentiment is then **compared across four dimensions**: overall corpus distribution, subreddit, temporal phase, and dominant frame. The cross-frame comparison provides a plausibility check for the frame classification: one would expect posts associated with replacement anxiety and workers' rights to display more negative sentiment than posts associated with innovation and opportunity.

4.4.2 Named Entity Recognition (NER)

While frame detection captures the interpretive lens through which AI is discussed, Named Entity Recognition (NER) is applied to identify the specific actors, institutions, and technologies that shape the discourse. This complementary perspective allows us to assess whether different communities not only framed AI differently, but centered their discourse on different institutional targets, and whether entity mentions co-occur systematically with specific frames.

The extraction uses **spaCy's transformer-based pipeline** “*en_core_web_trf*”, with a fallback to “*en_core_web_sm*” where computational constraints apply. The analysis focuses on **organizations**, **persons**, and **technology-related entities**, which are the categories most relevant to the research question.

Raw extraction from social media corpora produces considerable noise: web-scraping artifacts, ambiguous single-token abbreviations, and very low-frequency mentions that do not support meaningful analysis. An empirical **blacklist** is applied to remove these entries; the exclusion criteria are documented in the notebook. Organizationally relevant acronyms such as SAG-AFTRA and DGA are retained in full form where present and entities appearing fewer than twice in the corpus are excluded.

Entity distributions **are analyzed at two levels**: globally across the full corpus to identify the most prominent actors in the debate, and comparatively across subreddits and frames to assess whether different communities anchored their discourse to different institutional referents.

5. Results and Discussion

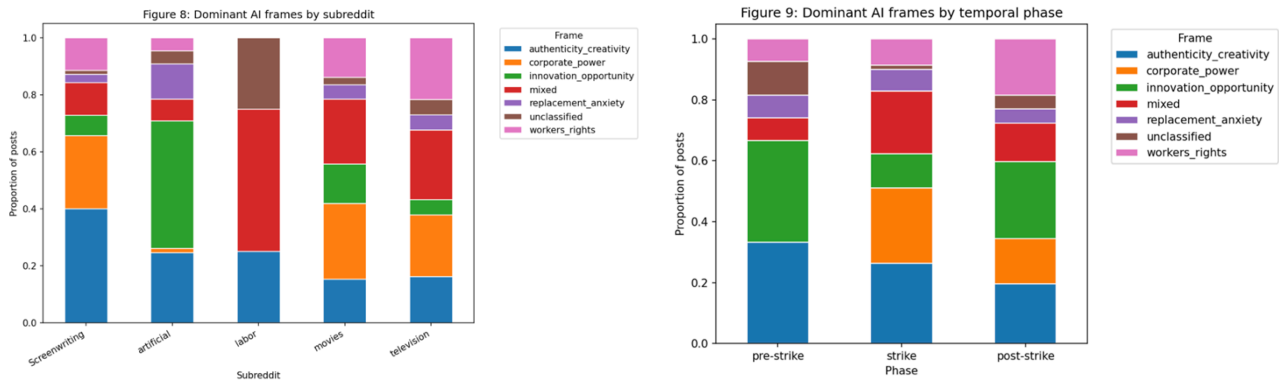
The analysis proceeds across four interconnected levels: frame detection, network structure, content analysis, and their integration.

5.1 Frame Detection

Frame detection reveals that AI discourse during the Hollywood Strike was not organized around a single narrative, but **around multiple interpretive orientations** that varied across communities and over time. At the corpus level, *Authenticity & Creativity* emerges as the most frequent dominant frame

(24.7%), followed by *Corporate Power* (18.8%) and *Innovation Opportunity* (18.4%). The relatively large share of mixed posts (16.5%) suggests that many discussions combined concerns about labor, creativity, and technological change rather than focusing on a single issue.

The cross-subreddit distribution shown in Figure 8 highlights substantial differences in framing patterns across communities. **r/Screenwriting** is strongly associated with *Authenticity & Creativity* (40.0%) and, to a lesser extent, *Corporate Power* (25.7%), a distribution consistent with the concerns typically discussed within the subreddit, where authorship, creative control, and the professional implications of AI are central topics. In contrast, **r/artificial** is dominated by *Innovation Opportunity* (44.6%), while *Corporate Power* is almost absent (1.5%), suggesting a community in which AI is more often discussed as a technological development than as a labor-related concern. The entertainment-oriented communities **r/movies** and **r/television** occupy an intermediate position, combining *Corporate Power* and *Authenticity & Creativity* in more balanced proportions.



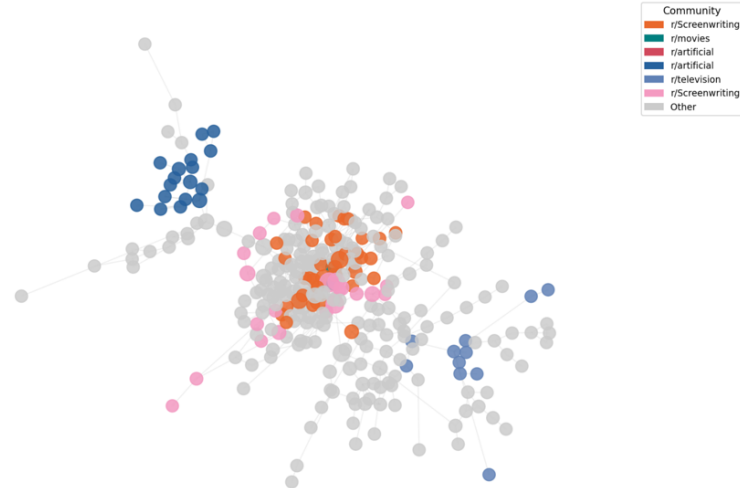
The temporal analysis shown in Figure 9 adds another dimension to these patterns. *Corporate Power* reaches its highest value during the strike phase (24.8%), consistent with the conflict-oriented nature of labor disputes. In the post-strike phase, *Innovation Opportunity* rises to 25.3%, suggesting a partial return of technology-focused discussions once the contractual negotiations were concluded. At the same time, *Authenticity & Creativity* declines from 33.3% in the pre-strike period to 19.5% post-strike, indicating that the discussion gradually shifted from questions of creative identity toward institutional and contractual concerns.

5.2 Network Analysis

The reply interaction network contains 2,684 nodes and 3,139 directed edges, with a density of 0.000436, a value typical of sparse online discussion networks. The largest weakly connected component includes 1,459 nodes (54.36% of the network), indicating that a **substantial proportion of users participated in isolated conversations** rather than in the main discussion space.

Figure 12b shows a network structure composed of visually distinct clusters. Combined with the community composition analysis, these clusters closely align with subreddit boundaries. This observation is confirmed by Louvain community detection, which identifies 33 communities with a global modularity of 0.867. Such a high modularity value indicates that interactions occur predominantly within communities rather than across them. The largest detected communities correspond almost entirely to individual subreddits, particularly **r/Screenwriting**, **r/artificial**, **r/movies**, and **r/television**.

Figure 12b: Reply network: top 300 users by betweenness centrality
Node size $\propto$ betweenness centrality · Color = Louvain community (top 6)



The broker analysis further clarifies the role of central users. Although the highest betweenness centrality value is observed for the user spikej (0.026), 9 of the 10 most central users are active in only one subreddit, and 8 belong primarily to r/Screenwriting. High betweenness centrality in this network therefore reflects **prominence within a single community** rather than genuine cross-community mediation. These users function primarily as conversational hubs within their own communities rather than as bridges connecting different parts of the network.

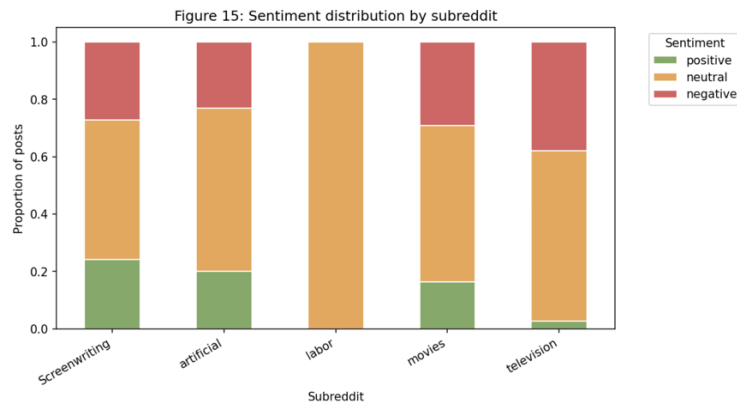
The temporal evolution of the network reveals increasing segmentation throughout the dispute. Modularity rises from 0.689 in the pre-strike period to 0.882 during the strike, before remaining high at 0.876 post-strike. Rather than encouraging cross-community interaction, the strike appears to have **reinforced existing discussion boundaries**. This interpretation is supported by cross-subreddit edge analysis: the proportion of interactions connecting users from different subreddits decreases from 15.5% pre-strike to 14.6% during the strike and 12.4% post-strike.

Degree assortativity remains negative throughout the observation period (-0.302 pre-strike, -0.225 during, -0.112 post-strike), confirming the **hub-and-spoke structure** commonly observed in online discussion networks. The gradual movement toward less negative values suggests the emergence of a more stable conversational core during the later stages of the debate.

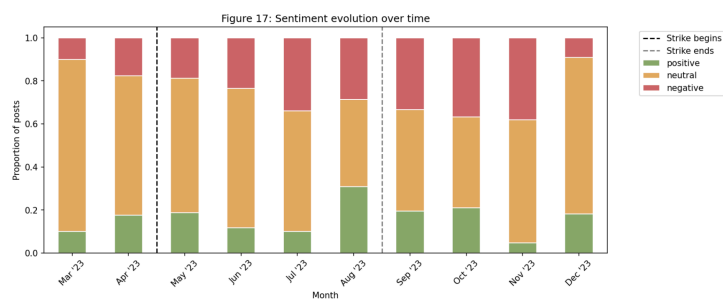
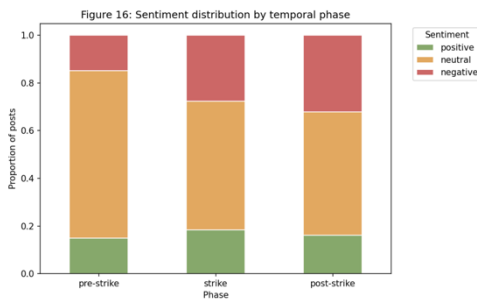
5.3 Content Analysis

Sentiment analysis assigns labels to all 255 posts. Neutral sentiment is the most common category (54.9%), followed by negative (27.8%) and positive (17.3%). The **predominance of neutral content** is consistent with the large number of informational and news-sharing posts present in the dataset. Among posts expressing a clear emotional orientation, negative sentiment is considerably more frequent than positive sentiment.

Figure 15 reveals meaningful variation across communities. r/television displays the highest proportion of negative posts. One possible explanation is that television audiences discussed AI primarily through concerns about the impact on writers, actors, and the future quality of television content, leaving relatively little space for technology-oriented or optimistic perspectives. The remaining communities show more balanced distributions. r/Screenwriting and r/movies occupy an intermediate position, combining substantial neutral and negative components. These patterns are broadly consistent with the frame analysis and suggest that communities differ not only in how they frame AI but also in the emotional tone associated with those discussions.

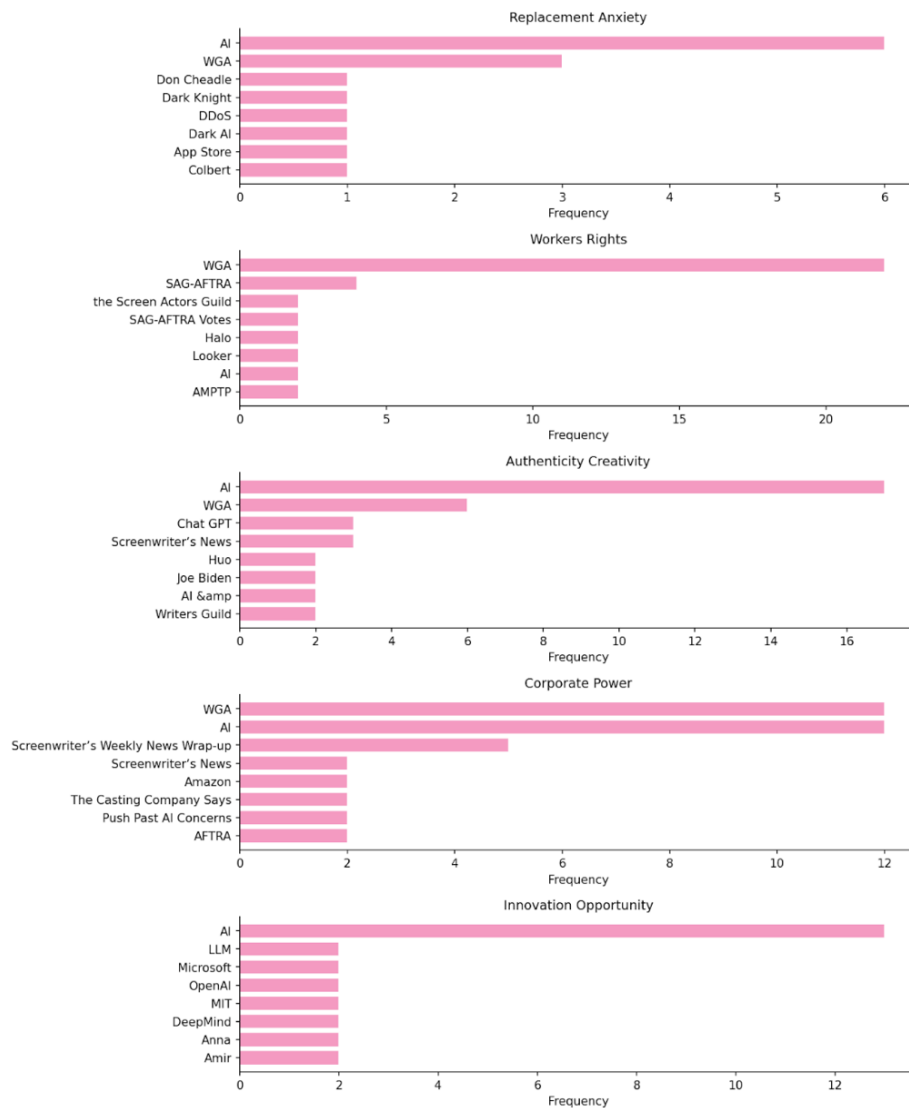


One notable result emerges from the temporal analysis shown in Figure 16. **Negative sentiment increases progressively** across the three phases and reaches its highest level in the post-strike period. Rather than disappearing after the formal resolution of the dispute, criticism remained visible within the discussion. One plausible interpretation is that attention shifted from mobilization against the threat of AI toward evaluation of the agreements that followed, including the adequacy of the protections negotiated by the unions. While the available data cannot establish causality, the result suggests that concerns about AI remained salient even after the dispute formally ended. Figure 17 provides a more fine-grained monthly view of sentiment evolution, showing that **the increase in negative sentiment was not linear but concentrated in specific months** of the strike and post-strike periods.



Named Entity Recognition provides a complementary perspective by linking framing patterns to the actors and institutions discussed within each frame. Applied to 204 posts, the analysis identifies **WGA and SAG-AFTRA among the most frequently mentioned entities**, alongside recurring references to **AI-related technologies and organizations** across the corpus. As shown in Figure 19b, entity distributions vary across frames, although some actors appear consistently throughout the debate. Workers' Rights is strongly associated with union-related entities such as WGA and SAG-AFTRA, whereas Innovation Opportunity includes technology-oriented entities such as OpenAI, Microsoft, and DeepMind. Authenticity & Creativity and Corporate Power display a more heterogeneous mix of creative, institutional, and technological actors. Thus, the results suggest that different frames tend to emphasize different sets of actors, providing additional support for the validity of the frame classification.

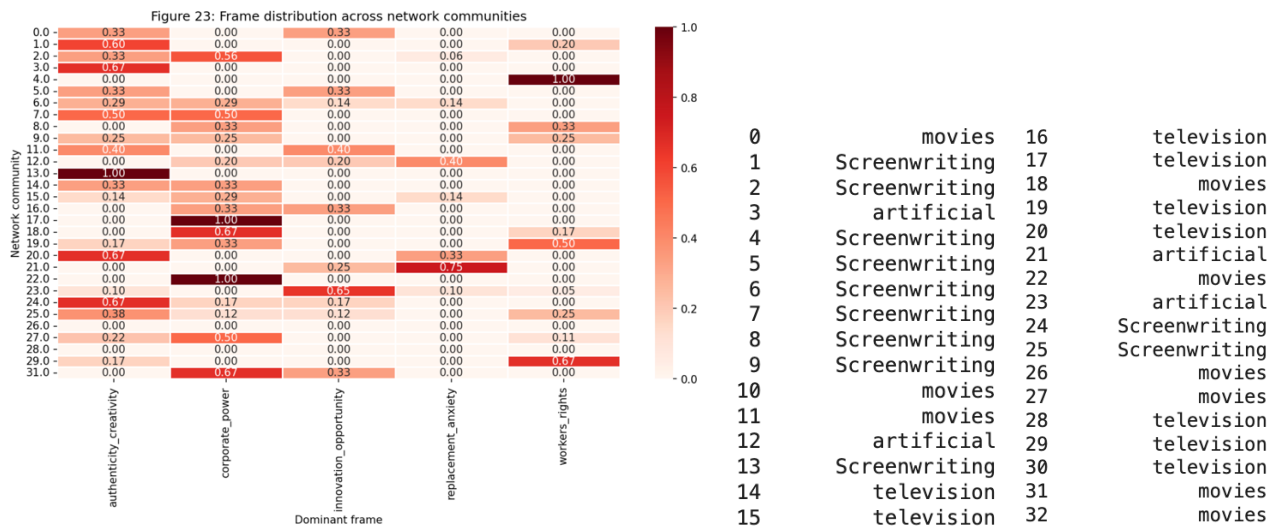
Figure 19b — Most mentioned entities by dominant frame



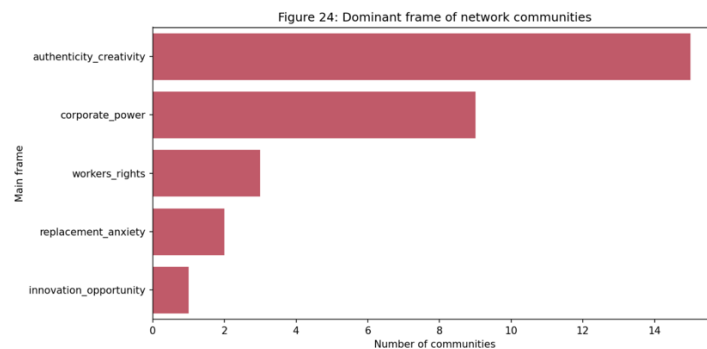
5.4 Integrated Community Analysis

The integration of network structure and content features provides the most direct answer to the research question. The analysis focuses on the 117 posts for which both community assignments and content labels are available.

Figure 23 displays the distribution of dominant frames across network communities. Although individual communities vary considerably in size, the heatmap suggests that **different interaction clusters tend to emphasize different aspects of the AI debate**. Communities dominated by r/Screenwriting are more frequently associated with *Authenticity and Creativity* or *Corporate Power*, whereas communities originating from r/artificial show a stronger presence of *Innovation and Opportunity* and *Replacement Anxiety*. Other communities, particularly those associated with entertainment-oriented subreddits, display a more heterogeneous framing profile, including including *Workers' Rights*, *Corporate Power*, and *Authenticity and Creativity*. Thus, the results suggest that structural communities are associated with different discursive orientations, even though the relationship is not perfectly deterministic.



This pattern is further reinforced by Figure 24, which summarizes the dominant frame of each community. *Authenticity & Creativity* emerges as the most common community-level orientation, followed by *Corporate Power*, while *Innovation and Opportunity* appears as the dominant frame in only a small number of communities.



Importantly, the network does not divide into a simple pro-AI versus anti-AI opposition. Instead, **communities become differentiated around concerns related to creativity, labor, corporate power, and technological opportunity.** Taken together, the results suggest that structural separation and discursive differentiation evolved in parallel. Communities that interacted less frequently also tended to emphasize different dominant frames, indicating that network boundaries and interpretive boundaries often reinforced one another throughout the debate.

6. Conclusion

This study examined how online communities framed the role of artificial intelligence during the 2023 Hollywood Strike through a combined analysis of content and interaction patterns on Reddit. Rather than producing a single, shared public debate, the discussion evolved into a set of parallel conversations shaped by the identities, interests, and concerns of different communities.

The findings indicate that AI was not interpreted primarily as a technology, but as a social issue whose meaning depended on the perspective of the community discussing it. Professional communities such as r/Screenwriting emphasized questions of authorship, creative autonomy, and labor protection, whereas technology-oriented communities such as r/artificial focused more heavily on innovation and

technological progress. Entertainment-focused communities occupied an intermediate position, combining concerns about corporate decision-making with broader audience perspectives. These differences were visible not only in the frames used to discuss AI, but also in sentiment patterns, entity references, and interaction structures.

Network analysis further showed that these interpretive differences emerged within a highly segmented discussion environment. Users interacted predominantly within their own communities, and cross-community interaction declined rather than increased over the course of the strike. As a result, communities that were structurally separated also tended to develop distinct discursive orientations. The debate therefore appears less as a shared public discussion and more as a collection of parallel conversations shaped by distinct community perspectives.

A particularly interesting finding concerns the temporal evolution of the discourse. While the strike itself intensified discussions around corporate power and labor conflict, the end of the dispute did not correspond to a reduction in concern about AI. Negative sentiment remained substantial and, in some cases, increased during the post-strike phase. This suggests that the controversy surrounding AI extended beyond the immediate labor dispute and reflected broader uncertainties about the governance of emerging technologies, the future of creative work, and the ability of institutions to regulate technological change effectively.

More broadly, the Hollywood Strike illustrates how public debates about AI are often debates about power, labor, and professional identity rather than about technology alone. The same technological development can be interpreted as an opportunity, a threat, a governance problem, or a challenge to creative authenticity depending on the social position of those affected by it. Understanding these differences is essential for interpreting public reactions to AI and for designing policies capable of addressing the concerns of multiple stakeholder groups.

Several limitations should be acknowledged. The dataset relies on Reddit submissions collected through the PullPush archive and therefore may not provide complete coverage of all relevant discussions. Frame detection is based on dictionary matching and cannot capture the full nuance of human interpretation. In addition, the pre-strike phase is represented by a smaller number of observations, limiting the robustness of some longitudinal comparisons.

Future research could extend the analysis beyond Reddit, compare different social media platforms, or examine whether similar framing dynamics emerge in other labor disputes involving artificial intelligence. Combining dictionary-based approaches with topic modeling or large language model-based classification could also provide a more detailed understanding of how AI-related narratives evolve over time.

Project materials are available in the accompanying [Google Drive repository](#).

AI Tools and Assistance Declaration

AI-based assistants (ChatGPT and Grammarly) were employed to improve the clarity, structure, and grammatical correctness of the documentation. ChatGPT was also used to suggest candidate items for predefined analytical components, including the query log, relevance keywords, custom stopwords, frame dictionaries, and entity blacklist, as well as for occasional debugging assistance. All methodological design choices, implementations, analyses, and interpretations were conceived, implemented, and validated by the authors.